

When Reputation Backfires: Social Pressure Can Reduce Cooperation in LLM Climate Societies

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Abstract

Reputation systems are widely assumed to stabilize cooperation. We ask whether that holds for large language model (LLM) agents in a climate-risk public-goods economy with asymmetric institutions. Twenty-six agents interact for thirty rounds under climatic shocks, a dual-use loss-and-damage pool, democratic rule updating, and social-information channels that include peer scoring, reputation, and gossip. Two independent replications share the full protocol and differ only by random seed. Contribution intensity relative to wealth falls, on average, after bad-reputation and gossip events among agents in the non-enforcing institution. Responses among enforcing-institution agents reverse sign across replications. Average cooperation remains moderately positive, yet democratic adaptation and wealth trajectories diverge. Social monitoring can coincide with withdrawal when agents lack peer-enforcement tools. The finding challenges the default claim that reputation raises cooperation in LLM climate societies.

Keywords: large language model agents; multi-agent systems; reputation; gossip; cooperation; climate risk sharing; public goods

1. Introduction

Collective climate action couples a public-goods dilemma with unequal vulnerability and contested redistribution. Agents must decide how much private wealth to transfer when peers can observe, gossip about, and sanction behavior, while formal rules may change through voting. Classical theory treats reputation as a cooperation technology: visible giving raises standing, and standing disciplines free-riding (Milinski et al. 2002; Wedekind and Milinski 2000; Nowak 2006). That claim now travels into societies of large language model (LLM) agents, where peer talk and scoring are easy to scaffold (Park et al. 2023; Piatti et al. 2024; Ren et al. 2025).

LLM agents also exhibit calculated free-riding once payoff frames dominate deliberation (Li and Shirado 2025; Guzman Piedrahita et al. 2025; Rand et al. 2012). Climate asymmetry sharpens the tension. Developing agents may lack peer-enforcement tools, so social stigma can land without a credible repair path. Forced institutional assignment removes exit as an escape valve that free-choice settings allow.

We ask whether negative reputation and gossip raise or lower subsequent contribution intensity for LLM agents under asymmetric institutions and climate risk. We embed Meta

Llama 3.1 8B agents in a repeated climate-risk economy with two parallel institutions, climatic shocks, a dual-use loss-and-damage fund, democratic whitelist voting, pairwise consistency scoring, aggregate reputation, and gossip bulletins. Contribution intensity relative to wealth is the primary outcome. We event-study intensity after bad-reputation marks, reputation drops, and gossip targeting, and we code post-event rationales for opportunistic versus repair motifs. Two independent replications share the protocol and differ only by random seed, so associations that recur can be separated from realization-specific trajectories.

Our contributions are fourfold. First, we provide an event study of reputation and gossip under forced climate institutions. Second, we show that agents without Stage-2 enforcement reduce intensity after social marks in both replications, while enforcing-institution responses reverse sign. Third, we show that positive average cooperation coexists with limited path stability and sparse reputation-repair language. Fourth, we document coverage without equalization in the loss-and-damage pool across both fiscal trajectories. Together, these results challenge the default claim that social monitoring raises cooperation in LLM climate societies.

2. Literature Review

2.1 Collective action, sanctions, and reputation

Collective-action problems arise when private incentives erode shared returns (Olson 1965; Hardin 1968). Taking an empirical lens, laboratory and evolutionary traditions show that monitoring, costly punishment, and reputation can sustain cooperation (Fehr and Gächter 2002; Boyd et al. 2003; Axelrod and Hamilton 1981; Axelrod 1984). More precisely, indirect reciprocity and image scoring formalize how observed giving shapes future partners (Wedekind and Milinski 2000; Panchanathan and Boyd 2004). Looking at this we can see that reputation helps solve commons dilemmas by linking contribution to standing and is among the core evolutionary routes to cooperation (Milinski et al. 2002; Nowak 2006). This is important because gossip can transmit reputation beyond direct observation (Sommerfeld et al. 2007). Across these lines, the working hypothesis is directional: more social visibility should raise, or at least protect, cooperative transfers.

Beyond informal sanctions, institutional design complements reputation and Ostrom documents polycentric rules, graduated sanctions, and participatory governance in durable commons regimes (Ostrom 1990). Those design principles

hence motivate dual channels in computational societies: formal enforcement inside one institution and social evaluation everywhere, with periodic voting on fiscal parameters. Yet we observe that classical literatures rarely ask what happens when social marks land on agents who cannot exit and cannot punish peers.

2.2 Language-model agents as strategic laboratories

Recent work operationalizes these dynamics in language-model agents, which produce numeric choices together with natural-language rationales, belief updates, and peer judgments (Park et al. 2023; Sumers et al. 2024). Consequently, game-theoretic LLM platforms treat such agents as laboratories for strategic interaction while cooperation benchmarks report heterogeneous success across games, prompts, and communication regimes (Mao et al. 2023; Piatti et al. 2024; Madmoun and Lahlou 2025; Liang et al. 2025). On balance some societies sustain contributions when talk is rich; others collapse once incentives dominate.

Interestingly enough, a growing strand shows that explicit reasoning can reduce cooperation. Specifically, models that deliberate more intensely often favor individually rational free-riding, mirroring human “spontaneous giving and calculated greed” (Li and Shirado 2025; Guzman Piedrahita et al. 2025; Rand et al. 2012). Relatedly, norm-focused multi-agent studies report partial emergence and fragile stability rather than automatic internalization (Gupta et al. 2025; Ashery et al. 2025; Cross et al. 2025). Likewise, human public-goods experiments show conditional cooperation and intuition–deliberation tensions that foreshadow these LLM patterns (Fischbacher et al. 2001; Rand et al. 2012). All in all, these results motivate separating average transfer levels from norm strength.

2.3 Reputation as an assumed stabilizer

Returning to reputation specifically, in several recent LLM-agent studies, reputation or peer talk are treated as stabilizers of cooperation. (Ren et al. 2025; Piatti et al. 2024; Madmoun and Lahlou 2025). Taking from classical evolutionary models, the punishment-strategy transplants show that explicit sanctions can organize behavior when costs and targets are clear (Warnakulasuriya et al. 2025). However, where gossip substitutes for formal punishment, the mapping from stigma to repair is unsettled. Compounding this uncertainty, self-reinforcing salience loops in LLM inference can lock opportunistic frames once they appear (Drake 2025). Prior LLM work rarely event-studies contribution intensity after negative social marks under forced institutional partitions.

2.4 Climate risk sharing and the research gap

Climate governance couples mitigation public goods with asymmetric vulnerability and compensation claims. Loss-and-damage instruments are dual-use in spirit: contributions can fund shared returns and contingent payouts after shocks. Computational societies that combine those fiscal rules with LLM agents that gossip and vote remain rare. Endogenous democracy lets agents revise punishment, subsidy, and equity weights, which is politically meaningful and confounds clean shock identification when votes coincide with damage rounds.

Consequently, prior work largely assumes that reputation and gossip raise cooperation, or at least do not systematically reduce it. LLM-agent papers that celebrate communication rarely measure post-stigma intensity under climate asymmetry across independent replications. We address that gap with an event study of contribution intensity after reputation and gossip events in a dual-replication climate-risk society.

3. Methodology

The environment is designed to place social pressure where classical theory expects it to help, while removing exit and equalizing enforcement rights that would otherwise confound the test. Agents inhabit a repeated public-goods economy with developed and developing roles that govern wealth, vulnerability, and institutional assignment. Each round, agents choose Stage-1 contributions inside their assigned institution. Only members of the enforcing institution then allocate Stage-2 sanctions and rewards. Climatic shocks, fund accounting, social scoring, and democracy occur on a fixed schedule described in Section 4.

3.1 Why two institutions

Cooperation problems often mix formal sanctions with informal evaluation. We therefore run two institutions in parallel. The enforcing institution (SI) combines Stage-1 public-goods returns with peer punishment and reward, framed to agents as a binding treaty. The non-enforcing institution (SFI) provides Stage-1 returns without Stage-2 sanctioning, framed as a non-binding agreement. In the climate package, assignment is forced every round: developed agents enter SI and developing agents enter SFI. The partition creates asymmetric sanction reach while holding the social-information stack common. Free institutional choice is reserved for boundary baselines discussed later.

3.2 Stage-1 public good and contribution intensity

Let c_i denote agent i ’s Stage-1 contribution and let G denote agents in the same institution. With multiplier m and group size $n = |G|$, each member receives

$$\text{share}_i = \frac{m \cdot C}{n}, \quad C = \sum_{j \in G} c_j.$$

The marginal per-capita return is m/n . Contribution budgets equal current wealth, floored at zero. Decision cards state budget bounds, group size, marginal return, and recent same-institution averages.

Absolute contributions differ by orders of magnitude across groups. We therefore analyze contribution intensity

$$\text{prop}_{i,t} = \frac{c_{i,t}}{w_{i,t}},$$

where $w_{i,t}$ is end-of-round wealth. All event studies use changes in prop so that social marks are comparable across wealth scales.

3.3 Stage-2 enforcement as a second-order public good

Only SI members enter Stage-2, with a sanction budget proportional to wealth. Punishment tokens reduce a target’s payoff at

a fixed effect-to-cost ratio; reward tokens raise payoffs at unit cost. Enforcement is costly to the sender, so provision of sanctions is itself a second-order public good. SFI members skip Stage-2 entirely. We track sanction burden as context for interpretation rather than as the primary estimand.

3.4 Dual-use loss-and-damage fund

Climate risk sharing requires a redistributive instrument that is not identical to the public good. Stage-1 contributions therefore act as dual-use deposits: each contributed unit also enters a collective loss-and-damage pool. Agents are told that contributions support both emissions-reduction framing and disaster payouts. They do not observe the live pool balance. After climatic shocks, developing agents may receive payouts subject to coverage caps and equity weights. Coverage is the ratio of aggregate payouts to aggregate gross damage in a shock round. High coverage can coexist with widening wealth gaps when collection and growth dominate disbursement.

3.5 Social information without mechanical force

Three linked channels create social pressure without forcing higher contributions. After contributions, evaluators score targets on a one-to-ten consistency scale. Agents observe an aggregate peer-trust rating derived from those scores. Bad reputation is defined as an aggregate score below four; a reputation drop is a decline of at least one point between rounds. Low pairwise scores generate gossip items injected into subsequent prompts, naming the observer, target, score, and a short comment. Capacity limits keep only a top set of items. Gossip targeting in the event study is reconstructed from eligibility conditions consistent with bulletin generation. Any reputation repair must arise from prompted choice.

3.6 Democracy as inexpensive meta-governance

Every five rounds, agents may propose and vote on a bounded whitelist of parameters, including subsidy fraction, punishment and reward effects, and loss-and-damage equity or damage weights. Majority outcomes update the live rule set. Democracy lets agents rewrite payoff consequences of enforcement and redistribution without paying Stage-2 costs. We analyze adopted rules as environmental context for contribution dynamics.

3.7 Agent interface

Agents are prompted language models with a default self-interested persona and soft climate-role guidance. Outputs include contribution amounts, sanction maps, proposals, votes, and rationales. Belief states update after rounds with trust labels. Fluent rationales are evidence of stated reasoning, not proof of internalized norms.

4. Experimental Design

4.1 Protocol and replications

Primary evidence comprises two independent full-protocol societies that differ only by random seed. Each uses Meta Llama 3.1 8B with temperature 0.5 and top $p = 0.95$ for ordinary decisions, lower temperatures for pairwise scoring

and voting, and a 4096-token context window. Each society contains 26 agents (12 developed, 14 developing) interacting for 30 rounds under forced institutional assignment. Public-good multiplier m equals 1.6. Climatic shocks occur in rounds 5 and 10 with severities 0.10 and 0.20. Democracy sessions occur every five rounds and therefore coincide with both shocks. Vulnerability is 0.30 for developed agents and 1.50 for developing agents. Stage-2 budgets in SI equal $\max(20, \lfloor 0.05 \times \text{wealth} \rfloor)$. Default punishment effect-to-cost is 3; reward effect-to-cost is 1. Gossip bulletins are eligible when pairwise scores are at most 7, with a capacity of five items.

Both replications enter every analysis with equal weight. Associations that share sign across replications are treated as recurring under this protocol. Associations that reverse sign are reported as heterogeneous. Agent identifiers are not treated as fixed personas across replications, because group labels reshuffle for a substantial subset of identifiers.

4.2 Outcomes and events

The primary outcome is $\text{prop}_{i,t}$. Immediate event-study changes use $\Delta \text{prop}_{i,t} = \text{prop}_{i,t+1} - \text{prop}_{i,t}$. Bad reputation, reputation drops, and gossip targets follow the definitions in Section 3. Diff-in-diff contrasts, when reported for one replication, compare affected agents with same-round co-institution controls without the focal event. Secondary outcomes include zero-contribution shares, round-mean trajectories, developed-developing wealth gaps, loss-and-damage coverage, adopted democratic rules, and coded motifs in post-event rationales. SI versus SFI mean comparisons are descriptive: institution membership is collinear with group label under forced routing.

4.3 Qualitative coding

For the first replication, contribution rationales in rounds immediately after bad-reputation or gossip events are coded with a multi-label codebook covering opportunistic, reputation-management, conformity, named-peer, future-round, and gossip-reference motifs. A second coder independently labels a stratified subsample for reliability. Lexical contrasts between SI and SFI contribution rationales are summarized with keyness.

4.4 Boundary baselines

Shock-free free-choice baselines from related abstract public-goods runs motivate the climate design. In those baselines, reputation can pull agents toward SI when exit is available, and greedy zero contribution dominates without existential risk. Those patterns are cited only as boundary conditions. They are not re-estimated under the forced climate protocol.

5. Results

5.1 Cooperation levels

Average contribution intensity is moderately positive in both replications, and SI and SFI means remain nearly identical within each society (Table 1). Run B sits above Run A for most of the horizon (Figure 1). Round-mean series are only weakly aligned across replications, and neither series shows a significant Mann-Kendall trend. Medians stay below means,

Table 1: Cooperation levels and fiscal outcomes across two independent replications. Behavioral rows summarize contribution intensity; fiscal rows summarize wealth and fund stocks. SI/SFI means are descriptive under forced assignment.

	Run A	Run B
<i>Behavior</i>		
Mean intensity (all)	0.293	0.365
SI / SFI mean	0.291 / 0.296	0.364 / 0.366
SI / SFI median	0.194 / 0.034	0.205 / 0.179
SI / SFI zero share	6.4% / 16.2%	11.1% / 22.4%
<i>Fiscal</i>		
Final wealth gap	2.15×10^8	5.83×10^8
Final LDF pool	4.34×10^9	1.30×10^{10}
Shock coverage (R5, R10)	0.768	0.768

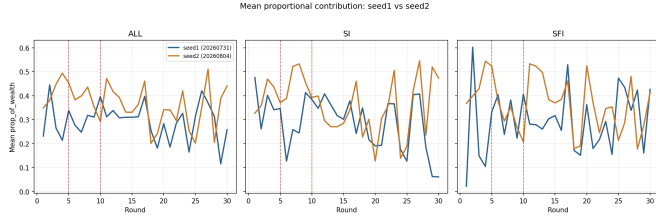


Figure 1: Round-mean contribution intensity by institution. Run B exhibits higher levels; within-run SI and SFI means stay close.

and zero-contribution shares remain higher in SFI than in SI. Matched institution means are descriptive under forced assignment rather than evidence of a causal institution effect.

Early-round composition differs. In Run A, most SFI agents contribute nothing in round 1, then escalate once peer history appears (Figure 2). In Run B, that opening freeze is largely absent. Both societies still settle into matched institution means after the opening window.

5.2 Social pressure and subsequent intensity

After gossip and bad-reputation events, mean next-round intensity falls among SFI agents in both replications (Table 2; Figure 3). Gossip associations are larger in magnitude than bad-reputation associations in SFI. Reputation-drop associations in SFI remain negative, though attenuated in Run B.

SI responses reverse sign across replications. In Run A, SI means after social marks are negative; in Run B they are positive, including a large positive gossip association. SI and SFI agents therefore behave differently under the same social-information stack. A universal claim that gossip lowers intensity for all agents is unsupported.

Figure 4 shows the Run A distribution of immediate changes. Mass lies below zero, especially for gossip targets in SFI. Table 3 contrasts marked agents with same-round controls in Run A. Diff-in-diff estimates are negative for all three event families, and permutation tests reject the null for bad reputation and gossip. Controls show near-zero or slightly positive mean changes, so the association concentrates among marked agents. First events coincide with larger declines than repeats in Run A.

Gossip naming skews sharply: a small subset of agents draws most of the bulletin mentions. Eligibility, by contrast, is widespread as most pairwise consistency scores fall at or below the gossip threshold. Named agents are often relatively high contributors at the hit.

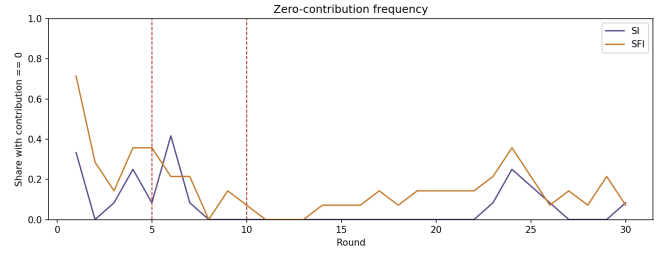


Figure 2: Zero-contribution shares by institution in Run A. SFI opens with widespread zeros; SI spikes after the first shock.

Table 2: Mean next-round change in contribution intensity after social marks. Negative SFI cells indicate average withdrawal; SI cells reverse sign across replications.

Event	Inst.	Run A	Run B
Bad reputation	SI	-0.036	+0.055
Bad reputation	SFI	-0.103	-0.069
Reputation drop	SI	-0.091	+0.058
Reputation drop	SFI	-0.119	-0.017
Gossip target	SI	-0.048	+0.187
Gossip target	SFI	-0.201	-0.084

5.3 Rationales after social events

Post-event rationales rarely describe reputation repair (Table 4). In Run A ($N=114$), opportunistic framing appears 29 times and reputation-management language only 3 times whereas explicit gossip references appear 0 times. Inter-coder agreement on a stratified subsample is substantial (percent agreement 0.83; Cohen's $\kappa=0.72$ for primary labels). Agents more often cite weak marginal returns than restored standing. Cross-replication keyness contrasts show lexical differences: SI texts emphasize strategy and self-interest; SFI texts emphasize incentives and immediacy. Fairness language remains rare in contribution talk.

5.4 Shocks, democracy, and redistribution

By Wilcoxon tests, shock-window mean intensity changes are statistically non-significant, yet composition still moves. A sharp SI zero spike follows the first shock in Run A, accompanied by payoff-maximizing rationales despite large remaining wealth. Within one or two rounds, mean intensity recovers. Because shock and democracy rounds coincide, climate damage and rule politics become entangled.

In both replications, democracy expands subsidies and equity themes, though the literal rule path diverges (Table 5). Punishment weakens in Run B alone. As for proposal rhetoric, it amounts to portable cooperation boilerplate.

In both replications, shock coverage equals about three-quarters, while terminal pools dwarf cumulative payouts and wealth gaps widen (Figures 5 and 6). Stock equalization, however, does not follow from high coverage. Since agents cannot observe the numeric pool balance, fund-stock optimization is an implausible behavioral account.

6. Discussion

Classical reputation theory predicts repair after social stigma (Milinski et al. 2002; Wedekind and Milinski 2000). The recurring association among SFI agents points in the opposite

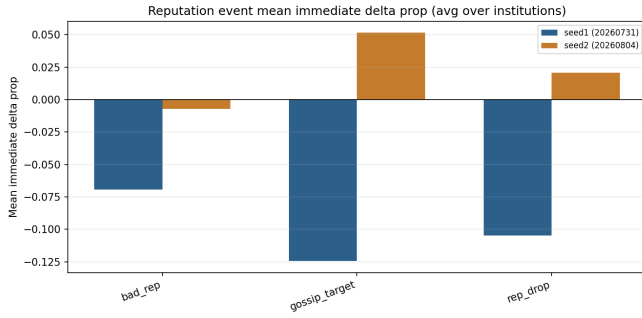


Figure 3: Mean Δprop after social marks by institution and replication. SFI cells stay negative; SI cells reverse sign.

Table 3: Diff-in-diff of Δprop versus same-round controls in Run A. Marked agents decline relative to unmarked peers in the same round.

Event	Aff. Δ	Ctrl Δ	DiD [95% CI]	d	p_{perm}
Bad rep.	-0.080	+0.014	-0.288 [-0.860, 0.203]	-0.10	0.015
Gossip	-0.139	+0.019	-0.208 [-0.601, 0.189]	-0.18	0.039
Rep. drop	-0.105	+0.028	-0.143 [-0.291, -0.017]	-0.15	0.072

direction. Why would social marks precede withdrawal rather than restored giving?

One mechanism is strategic resignation under asymmetric sanction reach. SFI agents can be named and scored, yet they cannot punish peers in Stage 2 and cannot exit into an enforcing institution. Once marked, the expected return to repairing standing may look low relative to conserving wealth, especially when marginal per-capita returns are weak. Withdrawal then becomes a constrained best response rather than a failure of understanding.

A second mechanism is payoff-frame lock-in. Post-event rationales emphasize marginal returns and self-interest far more often than credibility repair. That pattern fits calculated-greed accounts of deliberative LLM play (Li and Shirado 2025; Guzman Piedrahita et al. 2025; Rand et al. 2012). Social stigma may increase the salience of individual rationality without activating an image-scoring motive.

A third mechanism is noisy targeting. Gossip eligibility is nearly ambient because most pairwise scores fall below the bulletin threshold. Named agents are often relatively high contributors at the hit. If social marks frequently land on visible givers, agents may update toward “giving invites scrutiny” rather than “giving restores standing.” Belief labels also over-represent free-rider tags relative to cooperative tags, which can reinforce defensive conservation.

SI agents face a different incentive environment. They retain Stage-2 tools and higher absolute wealth, so reputation can still function as a coordination or insurance signal in some trajectories. The sign reversal across replications is consistent with history-dependent evolution under stochastic LLM sampling rather than with a single SI law of motion. Matched mean intensity between SI and SFI does not erase that difference in response to social marks.

Forced assignment matters for all three mechanisms. In free-choice baselines, high contributors can flee over-enforcement into SFI. Here, developed agents cannot exit SI and developing agents cannot acquire Stage-2 reach. Identification without

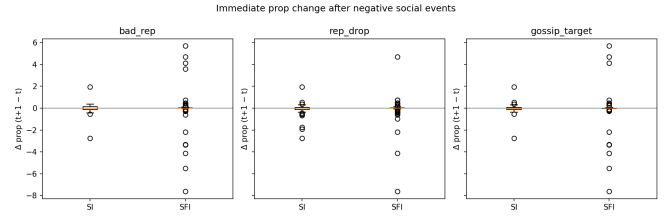


Figure 4: Distribution of immediate Δprop after social marks in Run A.

Table 4: Motifs in contribution rationales after bad-reputation or gossip events ($N=114$ texts in Run A). Opportunistic framing dominates reputation repair.

Motif	Count
Opportunistic / payoff-maximizing	29
Conformity	5
Reputation management / repair	3
Named peer references	3
Future-round planning	1
Explicit gossip reference	0

sanction reach, already visible in mixed baselines, reappears as stigma without repair.

Democratic adaptation and fiscal outcomes sharpen the interpretation. Both replications expand subsidies, yet only one weakens punishment, so political trajectories diverge under identical whitelists. Costless meta-governance can rewrite enforcement payoffs without building contribution norms. Identical shock coverage alongside widening wealth gaps shows that dual-use collection need not equalize stocks when the pool balance is hidden and absolute developed transfers dominate.

Observation and interpretation should stay separate. The data show a recurring negative association between social marks and next-round intensity among SFI agents, sparse repair language, and divergent SI and democratic trajectories. The mechanisms above are candidate accounts of that pattern. They are not uniquely identified by the present design.

7. Limitations

Several limits bound the claims. Only two replications and one open-weight 8B instruct model are studied. Additional replications are needed before SI gossip responses can be characterized distributionally. Forced institutional assignment precludes causal SI-versus-SFI identification. Shock and democracy rounds coincide. Gossip targeting is reconstructed from eligibility rules rather than from a complete verbatim bulletin archive. Agent identifiers are not stable personas across replications.

The environment is a stylized mechanism laboratory. It is not a calibrated model of Parties, pledges, or disbursement under the Fund for responding to Loss and Damage. Ostromian design principles are not fully implemented. Positive mean cooperation is not proof of internalized norms. Boundary free-choice baselines motivate the climate package but are not re-run under the same dual-replication protocol here. External validity beyond this prompt family, model, and parameter whitelist remains open.

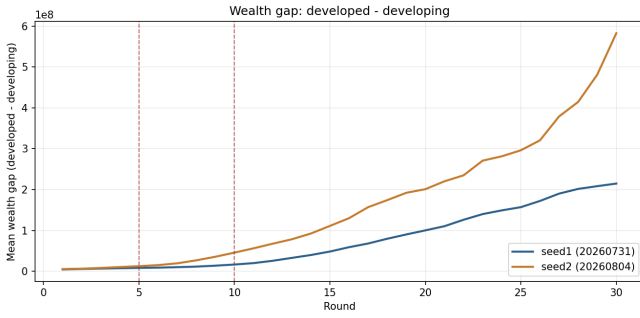


Figure 6: Developed-minus-developing mean wealth gap. Gaps widen despite matched coverage ratios.

Table 5: Adopted democratic rules. Both replications expand subsidies; only Run B weakens punishment.

Round	Run A	Run B
5	Subsidy fraction $\rightarrow$ 0.3	Subsidy top- $N \rightarrow$ 5
10	LDF equity weight $\rightarrow$ 0.5	Subsidy fraction $\rightarrow$ 0.5
15	Subsidy fraction $\rightarrow$ 0.4	LDF equity weight $\rightarrow$ 1.0
20	LDF damage weight $\rightarrow$ 1.5	Punishment effect $\rightarrow$ 2
25	LDF equity weight $\rightarrow$ 0.7	Subsidy fraction $\rightarrow$ 0.8
30	Subsidy fraction $\rightarrow$ 0.6	Subsidy fraction $\rightarrow$ 1.0

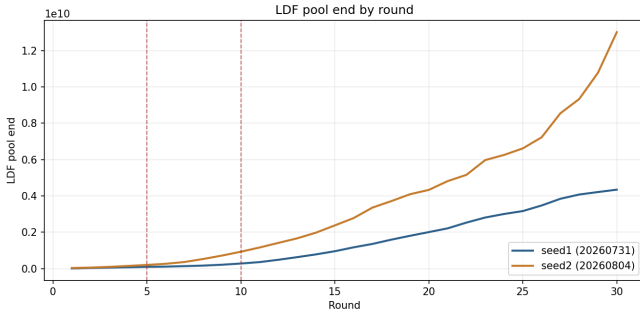


Figure 5: Loss-and-damage pool balances by round. Pools grow far beyond cumulative shock payouts.

8. Conclusion

Reputation and gossip are often assumed to stabilize cooperation in LLM agent societies. Under climate risk and forced institutional asymmetry, that assumption fails for developing agents in the non-enforcing institution. Across two independent replications, social marks associate with lower next-round contribution intensity in SFI. Responses among enforcing-institution agents reverse sign. Average cooperation remains moderately positive, democratic adaptation is history dependent, and high shock coverage coexists with widening wealth gaps.

The central finding is role-contingent social pressure. Monitoring without enforceable repair can coincide with withdrawal rather than with restored giving. Future work should expand replications and models, desynchronize shocks from democracy, restore free institutional choice under climate prompts, and ablate gossip thresholds. Those extensions can test whether restoring exit or Stage-2 reach converts social stigma back into cooperation technology.

References

Ashery, A. F., Aiello, L. M., and Baronchelli, A. (2025). Emergent social conventions and collective bias in LLM populations. *arXiv*

preprint arXiv:2410.08948.

Axelrod, R. (1984). *The Evolution of Cooperation*. Basic Books, New York.

Axelrod, R. and Hamilton, W. D. (1981). The evolution of cooperation. *Science*, 211(4489):1390–1396.

Boyd, R., Gintis, H., Bowles, S., and Richerson, P. J. (2003). The evolution of altruistic punishment. *Proceedings of the National Academy of Sciences*, 100(6):3531–3535.

Cross, L., Haber, N., and Yamins, D. L. K. (2025). Validating generative agent-based models of social norm enforcement: From replication to novel predictions. *arXiv preprint arXiv:2507.22049*.

Drake, S. (2025). Neural howlround in large language models: A self-reinforcing bias phenomenon, and a dynamic attenuation solution. *arXiv preprint arXiv:2504.07992*.

Fehr, E. and Gächter, S. (2002). Altruistic punishment in humans. *Nature*, 415(6868):137–140.

Fischbacher, U., Gächter, S., and Fehr, E. (2001). Are people conditionally cooperative? Evidence from a public goods experiment. *Economics Letters*, 71(3):397–404.

Gupta, P., Zhong, Q., Yakura, H., Eisenmann, T., and Rahwan, I. (2025). The role of social learning and collective norm formation in fostering cooperation in LLM multi-agent systems. *arXiv preprint arXiv:2510.14401*.

Guzman Piedrahita, D., Yang, Y., Sachan, M., Ramponi, G., Schölkopf, B., and Jin, Z. (2025). Corrupted by reasoning: Reasoning language models become free-riders in public goods games. *arXiv preprint arXiv:2506.23276*.

Hardin, G. (1968). The tragedy of the commons. *Science*, 162(3859):1243–1248.

Li, Y. and Shirado, H. (2025). Spontaneous giving and calculated greed in language models. *arXiv preprint arXiv:2502.17720*.

Liang, Y., Qu, Y., Yang, J., Lin, S., and Shen, Z.-J. M. (2025). Everyone contributes! incentivizing strategic cooperation in multi-LLM systems via sequential public goods games. *arXiv preprint arXiv:2508.02076*.

Madmoun, H. and Lahlou, S. (2025). Communication enables cooperation in LLM agents: A comparison with curriculum-based approaches. *arXiv preprint arXiv:2510.05748*.

Mao, S., Cai, Y., Xia, Y., Wu, W., Wang, X., Wang, F., Ge, T., and Wei, F. (2023). ALYMPICS: LLM agents meet game theory — exploring strategic decision-making with AI agents. *arXiv preprint arXiv:2311.03220*.

Milinski, M., Semmann, D., and Krambeck, H.-J. (2002). Reputation helps solve the ‘tragedy of the commons’. *Nature*, 415(6870):424–426.

Nowak, M. A. (2006). Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563.

Olson, M. (1965). *The Logic of Collective Action*. Harvard University Press, Cambridge, MA.

- Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge.
- Panchanathan, K. and Boyd, R. (2004). Indirect reciprocity can stabilize cooperation without the second-order free rider problem. *Nature*, 432(7017):499–502.
- Park, J. S., O’Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*.
- Piatti, G., Jin, Z., Kleiman-Weiner, M., Schölkopf, B., Sachan, M., and Mihalcea, R. (2024). Cooperate or collapse: Emergence of sustainable cooperation in a society of LLM agents. *arXiv preprint arXiv:2404.16698*.
- Rand, D. G., Greene, J. D., and Nowak, M. A. (2012). Spontaneous giving and calculated greed. *Nature*, 489(7416):427–430.
- Ren, S., Fu, W., Zou, X., Shen, C., Cai, Y., Chu, C., Wang, Z., and Hu, S. (2025). Reputation as a solution to cooperation collapse in LLM-based MASs. *arXiv preprint arXiv:2505.05029*.
- Sommerfeld, R. D., Krambeck, H.-J., Semmann, D., and Milinski, M. (2007). Gossip as an alternative for direct observation in games of indirect reciprocity. *Proceedings of the National Academy of Sciences*, 104(44):17435–17440.
- Sumers, T. R., Yao, S., Narasimhan, K., and Griffiths, T. L. (2024). Cognitive architectures for language agents. *Transactions on Machine Learning Research*.
- Warnakulasuriya, K., Dissanayake, P., De Silva, N., Cranefield, S., Savarimuthu, B. T. R., Ranathunga, S., and de Silva, N. (2025). Evolution of cooperation in LLM-agent societies: A preliminary study using different punishment strategies. *arXiv preprint arXiv:2504.19487*.
- Wedekind, C. and Milinski, M. (2000). Cooperation through image scoring in humans. *Science*, 288(5467):850–852.

A. Appendix: Protocol, Architecture, and Supplementary Analyses

This appendix supports replication of the scientific protocol and visualizes module interactions. Software-engineering artifacts are omitted.

A.1 Experimental configuration

Table 6 lists core configuration values shared by both climate-risk societies.

Table 6: Shared configuration for the two full-protocol replications.

Parameter	Value
Population	26 agents (12 developed, 14 developing)
Horizon	30 rounds
Institutional assignment	Forced: developed $\rightarrow$ SI; developing $\rightarrow$ SFI
Public-good multiplier m	1.6
Stage-1 budget	Current wealth (non-negative)
Stage-2 budget (SI)	$\max(20, \lfloor 0.05 \times \text{wealth} \rfloor)$
Punishment effect / cost (default)	3 / 1
Reward effect / cost	1 / 1
Initial subsidy fraction	0.20 of punishment costs to top contributors
Shock schedule	Round 5 severity 0.10; round 10 severity 0.20
Vulnerability	Developed 0.30; developing 1.50
Damage scale factor	$150000 \times \text{severity} \times \text{vulnerability}$
LDF deposit rule	Stage-1 contribution dual-use deposit each round
LDF recipients	Developing agents after shocks
Initial max coverage	0.90
Initial equity weight	0.00 (democratically revisable)
Democracy interval	Every 5 rounds
Consistency / gossip	Enabled; bulletin if score ≤ 7 ; max 5 items
Decision model	Meta Llama 3.1 8B
Default sampling	temperature 0.5, top $p = 0.95$
ToM / democracy temps	0.3 / proposal 0.5, vote 0.3
Context length	4096 tokens
Independent replications	Two random seeds (Run A, Run B)

A.2 Round protocol

Each round proceeds as follows.

1. Institutional assignment is applied.
2. Agents choose Stage-1 contributions under decision cards, climate role guidance, beliefs, and any gossip bulletins.
3. Stage-1 public-good returns are distributed within each institution.
4. Dual-use deposits update the loss-and-damage pool.
5. SI agents allocate Stage-2 punishments and rewards.
6. Pairwise consistency scoring updates peer judgments and aggregate reputation.
7. Eligible low scores generate gossip bulletins for later rounds.
8. If the round index is divisible by five, agents propose and vote on whitelist parameters.
9. If the round is a shock round, damages and developing-country payouts are realized.
10. Wealth, beliefs, and histories update for the next round.

A.3 Architecture: module interaction map

Figure 7 shows the intra-round control flow. Figure 8 shows cross-round information feedback into the next contribution prompt. Figure 9 distinguishes architectural writes from semantic reads, including the hidden loss-and-damage pool balance. Figure 10 isolates the dual-use Stage-1 contribution.

Canonical narrative flow for an agent is: observation $\rightarrow$ reasoning $\rightarrow$ contribution $\rightarrow$ reputation $\rightarrow$ gossip $\rightarrow$ proposal $\rightarrow$ vote $\rightarrow$ parameter change $\rightarrow$ later contribution. In climate mode, democratic “institutional change” means parameter change, not free switching between SI and SFI.

A.4 Prompt templates (sanitized)

A.4.1 Default behavioral persona

Act in whatever way maximizes your own cumulative payoff over the game. Cooperation, free-riding, punishment, or defection are all valid if they serve your interests. Do not assume you have any obligation to the group. Treat other agents as strategic variables.

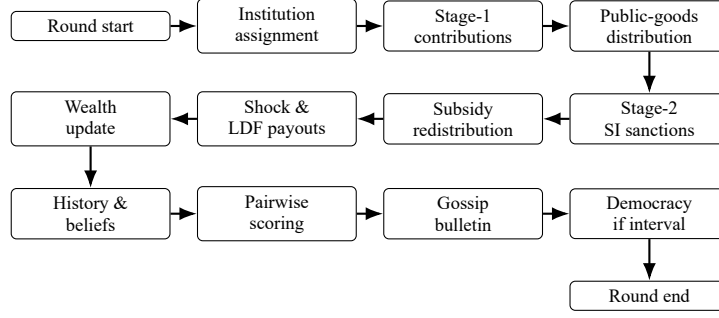


Figure 7: Intra-round module pipeline. Democracy applies only on interval rounds; otherwise the society proceeds directly to round end.

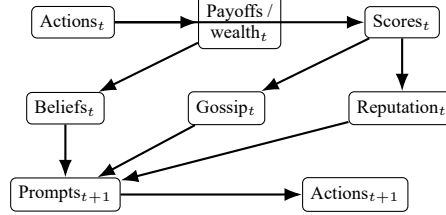


Figure 8: Cross-round feedback. Reputation and gossip from round t enter prompts in round $t + 1$. The numeric loss-and-damage pool balance is not fed into prompts.

Use past rounds to identify patterns you can exploit or respond to. Keep choices valid under all game rules and cite the concrete facts driving your decision.

A.4.2 Climate role guidance

Developed country. You typically have higher fiscal capacity and historically higher emissions. Treat long-run climate stability and institutional credibility as strategically important. You are assigned to the binding treaty institution with peer sanctioning.

Developing country. You typically face higher climate vulnerability and tighter fiscal constraints. Treat disaster compensation and survival constraints as strategically important. You are assigned to the non-binding agreement without peer sanctioning.

A.4.3 Stage-1 contribution decision

Choose an integer contribution between 0 and your current budget. You observe group size, marginal per-capita return, recent same-institution averages, own wealth, beliefs, and any gossip bulletins. State a brief rationale for the chosen amount.

A.4.4 Stage-2 sanction decision (SI only)

Allocate your sanction budget across listed peers as punishments and rewards. Punishments and rewards must be non-negative integers that respect the budget. Provide a brief justification that is consistent with the numeric allocation.

A.4.5 Pairwise consistency scoring

Score the target from 1 to 10 for consistency between stated intent and observed action. Provide a one-sentence justification. Low scores may generate gossip bulletins.

A.4.6 Democracy proposal and vote

Propose at most one whitelist parameter change, or abstain. Later, vote yes or no on each eligible proposal. Whitelist items include subsidy fraction, subsidy breadth, punishment effect, reward effect, and loss-and-damage equity or damage weights.

A.5 Supplementary statistics

Table 7 reports cross-replication correlations of round-mean intensity. Table 8 reports Wilcoxon tests around shock rounds. Figure 11 shows shock-delta distributions. Figure 12 shows agent-mean intensity scatter across replications.

Table 7: Cross-replication correlation of round-mean contribution intensity.

Series	Pearson r	p	RMSE
All agents	0.194	0.305	0.124
SI only	0.144	0.448	0.164
SFI only	0.020	0.916	0.184

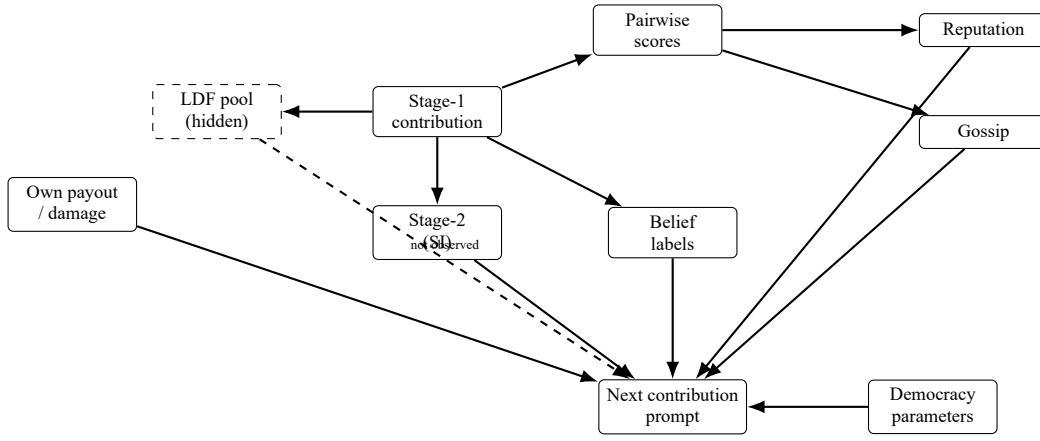


Figure 9: Semantic coupling among modules. Solid arrows are information channels available to agents. The dashed edge marks an architectural write without a semantic read: the live pool balance is hidden.

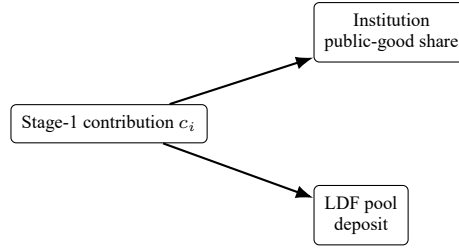


Figure 10: Dual-use Stage-1 contribution. One decision funds both institutional public-good returns and the loss-and-damage pool.

A.6 Motif codebook

- *Opportunistic*: free-riding, self-interest, payoff maximization, or marginal-return language.
- *Conformity*: peer or group-average behavior as the reason for the agent's own choice.
- *Reputation management*: explicit standing, credibility, trust scores, or avoiding a free-rider label.
- *Named agents*: reference to a specific peer identifier.
- *Future-rounds*: intertemporal conservation without a stronger primary motif.
- *Gossip reference*: explicit gossip or bulletin language.

A.7 Claim boundaries

Supported under the present protocol: recurring SFI withdrawal after gossip and bad reputation; within-run $SI \approx SFI$ mean intensity; coverage without equalization; history-dependent democracy. Unsupported: causal SI-versus-SFI institution effects; universal SI gossip withdrawal; evaluation of any real-world loss-and-damage fund; full implementation of Ostrom design principles; persona-stable agent identifiers across replications.

Table 8: Within-agent shock event-study Wilcoxon tests for Δprop .

Run	Shock	Mean Δ	Median Δ	Wilcoxon p
A	R5	+0.023	−0.014	0.367
A	R10	+0.010	+0.015	0.328
B	R5	−0.080	−0.011	0.582
B	R10	+0.050	−0.147	0.367



Figure 11: Within-agent post-minus-pre intensity changes around climatic shocks.

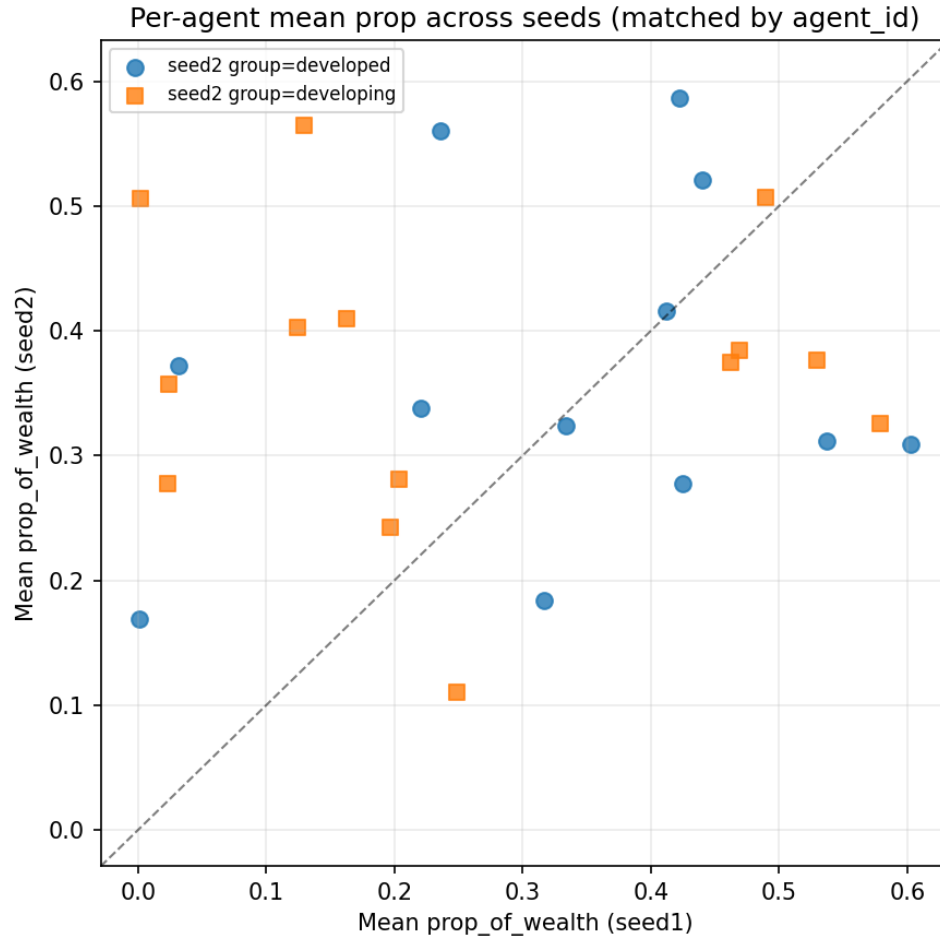


Figure 12: Agent-mean contribution intensity in Run A versus Run B. Points are not persona-matched across replications.